Wireless Network Setup and Security

1. Introduction

Wireless networks are widely used in homes, offices, and organizations to provide internet and network connectivity without physical cables. However, wireless networks must be configured properly and secured to prevent unauthorized access and network attacks.

This project demonstrates how to set up a basic wireless network using **Cisco Packet Tracer**. It includes configuring a router, switch, wireless router, and connecting wired and wireless devices while implementing basic wireless security measures.

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2. Objective

The objectives of this project are:

- To understand the basic setup of a wireless network.
- To configure routers and switches in a network environment.
- To connect wired and wireless devices.
- To configure wireless access using SSID and passphrase.
- To implement basic wireless security to prevent unauthorized access.

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3. Tools and Software Used

Tool	Purpose
Cisco Packet Tracer	Network simulation and configuration
Router 2911	Network routing
Switch 2960	Connecting multiple devices
Wireless Router WRT300N	Providing wireless connectivity
PC	Wired network device
Laptop	Wireless network device
Smartphone	Wireless network device

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4. Network Devices Used

Device	Model	Purpose
Router	2911	Main gateway of the network
Switch	2960	Connects wired and wireless devices
Wireless Router	WRT300N	Provides WiFi connectivity

Device	Model	Purpose
PC	End Device	Wired connection
Laptop	End Device	Wireless connection
Smartphone	End Device	Wireless connection

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5. Network Topology

The network topology consists of a router connected to a switch. The switch connects to a wired PC and a wireless router. The wireless router provides WiFi connectivity to the laptop and smartphone.

Network Structure:

```
Router (2911)
     |
   Switch (2960)
   /        \
PC          Wireless Router (WRT300N)
              /           \
          Laptop       Smartphone
```

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6. IP Address Configuration

Device	IP Address	Subnet Mask	Gateway
Router	192.168.1.1	255.255.255.0	-
Wireless Router	192.168.1.2	255.255.255.0	192.168.1.1
PC	192.168.1.10	255.255.255.0	192.168.1.2
Laptop	192.168.1.101	255.255.255.0	192.168.1.2
Smartphone	192.168.1.30	255.255.255.0	192.168.1.2

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7. Network Implementation Steps

Step 1: Adding Devices

The following devices were added in **Cisco Packet Tracer**:

- Router 2911
- Switch 2960
- Wireless Router WRT300N
- PC
- Laptop
- Smartphone

Step 2: Connecting Devices

The devices were connected using copper straight-through cables.

Connections made:

- Router connected to Switch
- PC connected to Switch
- Wireless Router connected to Switch
- Laptop and Smartphone connected wirelessly

Step 3: Router Configuration

Router CLI configuration:

```
enable
configure terminal
interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

This router acts as the main gateway of the network.

Step 4: PC Configuration

The PC connected to the switch was configured manually.

```
IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.2
```

Step 5: Wireless Router Configuration

Wireless router settings:

```
IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0
Gateway: 192.168.1.1
```

The wireless router was connected to the switch through its Ethernet port.

Step 6: Wireless Network Setup

Wireless network was configured with the following settings:

```
SSID: ShivamWiFi
Security Mode: WPA2 Personal
Passphrase: shivam123
```

This allows wireless devices to connect securely.

Step 7: Laptop Connection

The laptop connected to the wireless network using DHCP.

```
Assigned IP Address: 192.168.1.101
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.2
```

Step 8: Smartphone Connection

The smartphone connected to the wireless router using SSID and passphrase.

Manual configuration:

```
IP Address: 192.168.1.30
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.2
```

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8. Network Testing

The network connectivity was tested using the **ping command**.

Example:

```
ping 192.168.1.101
```

Result:

```
Reply from 192.168.1.101
```

This confirms that communication between devices is successful.

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9. Security Measures Implemented

The following security measures were applied:

- WPA2 Personal wireless encryption
- Secure WiFi passphrase
- Controlled access to wireless network using SSID

These measures help protect the wireless network from unauthorized access.

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10. Conclusion

This project successfully demonstrated the setup and configuration of a wireless network using **Cisco Packet Tracer**. Wired and wireless devices were connected through a router, switch, and wireless router. Wireless security was implemented using WPA2 encryption and passphrase protection. The network was tested successfully using ping commands, confirming proper communication between devices.